

The population means of the volumes of liquid in the bottles filled by machines A and B are denoted by μ_A and μ_B .

- (a)** Test at the 2% significance level whether there is any difference between μ_A and μ_B . [8]

[illegible]

- (b) Find the set of values of α for which there would be evidence at the $\alpha\%$ significance level that $\mu_A - \mu_B$ is greater than 0.25. [4]

[illegible]